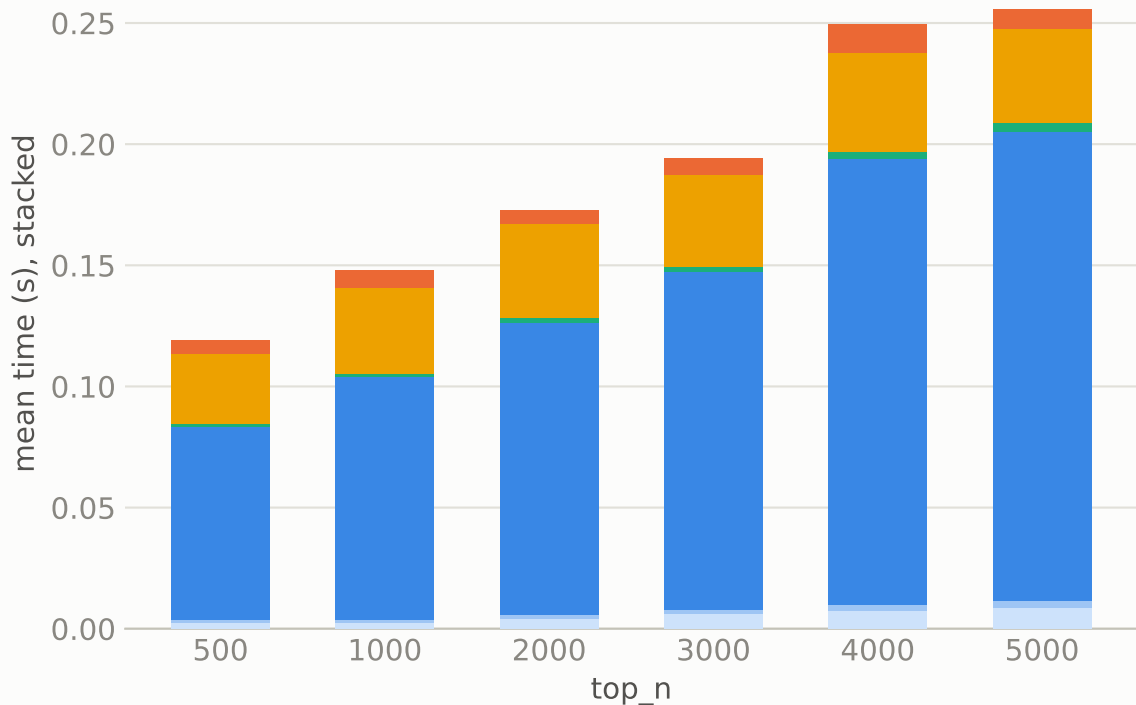


tier_10k: stage-time breakdown vs. top_n ($\sigma=0.30$, mean over α, k)



Probe (HNSW + overlap)
Union combine
 N_i/n_i lookup

Stage 1 (Dinkelbach)
Unionability scoring
Stage 2 (ILP)